AI PRODUCT MANAGEMENT CASE STUDY · WEEK ONE DONE, WEEK TWO UNDER WAY · NOVEMBER 2026

Clarivi

A plain-words morning answer from my own Apple Watch data

For Apple Watch users who train and sleep with the watch on. It says what last night's readings mean against your own normal, why the day's status was set, and one thing to do today. I built it and tested it on myself (I wrote the specs and decisions; Claude Code wrote the code to them), and I report what missed next to what worked.

The problem

Apple Watch records the numbers, and the Health app shows each as an isolated daily card. It does not say whether today's reading is unusual for you, or why.

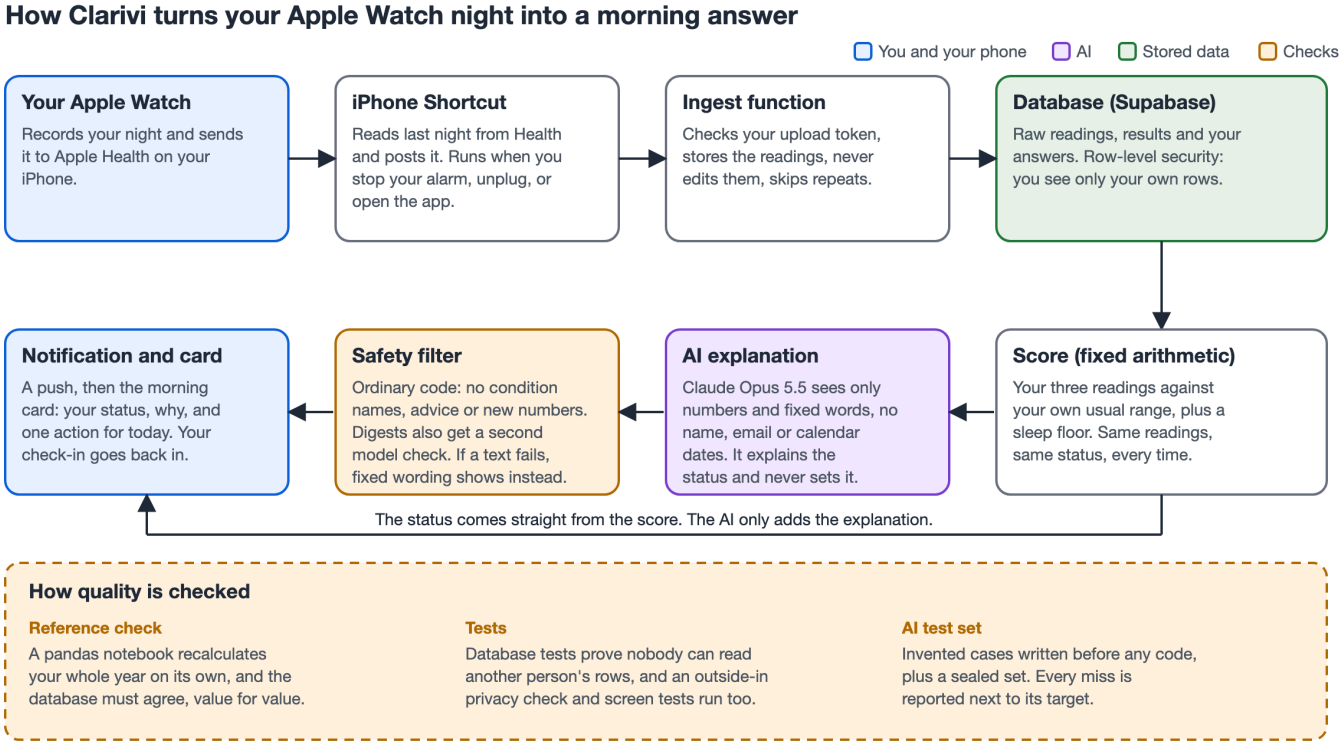
Clarivi explains them, in plain words and against the user's own normal, which Apple's own features do not.

261 nights
My year's nights, usual ranges and daily status agree with an independent check: 0 values differ. Same author, so independent code, not an independent person.

38 of 40
Invented cases passed the AI's safety filter on one live attempt. The filter was tuned on these cases.

4 of 7
Days with a sync in week one, against a target of 6: missed. Only 2 of 7 had a status before noon.

2 of 13
Disrupted mornings the score flagged, against a target of 2 in 3: missed. One person, thin evidence.



Readings flow in from an iPhone Shortcut, are scored on the server, and the AI only writes the words.

What I built

- **The card:** Ready, Ease off or Rest in a few plain sentences, one action for the day, and a "Why today" screen that shows how it was decided.
- **The score:** fixed arithmetic in SQL against my own last 42 nights, checked by separate pandas code. A night with sleep far below my usual cannot be Ready.
- **The AI** (added after week one): Claude Opus 5.5 writes the wording and never sets the status. It sees an allow-list of results only, and its text goes through a filter, a retry and a fixed fallback.
- **The sync:** an iPhone Shortcut on three triggers (alarm stopped, charger unplugged, app opened), with an 11:30 reminder if the status is missing.
- **Private by default:** row-level security on every table, no third-party scripts, three separate consent boxes.

Key decisions

- **A fixed score, not a learned model:** every status is explainable and checkable. The cost: it cannot adapt to me.
- **The AI explains, never decides:** a wrong sentence cannot change what the card says. The cost: a short wait, and a US AI provider that needs its own consent.
- **A smaller test, set before any result:** four people for 28 days became one person for a week. The cost: it cannot speak for anyone else.
- **Test before building:** I tried three planned analysis modules on my own year first. Two found nothing the data could support (0 of 12 each) and the third missed its target, so I built none.

How I measured quality

Ten targets, set on 29 September 2026, before any result. The six that could be counted; week one, 19 to 25 October:

Measure	Target	Result
Sync	6 of 7 days	**4 of 7**
Correctness	Match the reference	**0 of 261 nights differ**
Isolation	No tolerance	**Held**
Signal, my year	2 in 3	**2 of 13**
Quiet by default, my year	About 1 in 7	**52 of 231 days (1 in 4.4)**
Engagement	5 of 7 days	**3 of 7**

The other four (notifications, check-ins, action and my own account) are in docs/evaluation.md: two days with a status were too few to count the first three, and my own account is under What I learned.

What I learned

- The sync miss was partly the Watch: its sleep can reach Health hours after waking. A fixed wait failed on its first real morning, so I dropped it and made the reminder cover an unfinished night.
- A Ready on a very short night led to a sleep floor. It made the score louder than my "quiet" target (52 of 231 days, against 29 of 231 before it), and I report that as a miss.
- Counting the sync three ways showed that "4 of 7" counted any post, so week two counts the status shown before noon.
- My own account: it never changed a decision, and I would not keep using it as it was. Both point at the sync, which week two tests.

> **Limits:** one person, one week so far. The same author built the system and counted the results. Week two is not included yet.
> **Next:** week two (2 to 8 November) with the alarm trigger and the AI explanation live, counted as the status shown before noon; then a second real user.

In the repository (github.com/sabinejoseph8/Apple-Health): README, docs/evaluation.md, Project Memory (specs and plan), the architecture diagram and the checks. Demo on made-up data: apple-health-rust.vercel.app.

Skills: AI product management (specs, phased plan, success targets set first, evaluation, iteration from data), responsible AI and data protection.